

Name _____ Date _____ Period _____

Domain and Range: All Three Notations

Algebra II Honors | Unit 1 | Day 1 | TEK A2.7.I

Objective: I can write domain and range in verbal, inequality, set, and interval notation, determine a domain algebraically from a function rule, and use union notation for a domain that comes in separate pieces.

Vocabulary

Domain — the set of all _____ values, the _____-values. **Range** — the set of all _____ values.

Inequality notation — writes domain or range using _____ instead of interval or set notation.

Set-builder notation — $\{x \mid \text{condition}\}$, read "the set of all x _____ the condition holds."

Interval notation — _____ includes an endpoint, _____ excludes it. Infinity always takes a _____.

Union $\cup$ — joins two separate intervals when the domain has a _____ in it.

Undefined — an input the rule cannot accept. Two causes in this course: a _____ and a _____.

Part 1 — From Pairs and Tables

List least to greatest, each value once.

a. $\{(-4, -6), (0, -1), (2, 4), (3, -7), (7, 7)\}$ D _____ R _____

b. $\{(-5, 3), (-1, 0), (2, 3), (6, -8)\}$ D _____ R _____

c. Why does the range in (b) have three values when there are four pairs? _____

Part 2 — All Three Notations

Complete every cell.

Verbal description	Inequality	Set notation	Interval notation
all real numbers	_____	_____	_____
all reals at least -6	_____	_____	_____
all reals less than 5	_____	_____	_____
all reals from -3 to 3, 3 included only	_____	_____	_____
all reals except 2	_____	_____	_____
just the numbers -2, 0, and 4	(none)	_____	_____

The last two rows are the ones to think hardest about. Explain the difference between them:

Row 5 is continuous with one point removed; row 6 is _____

Part 3 — Finding a Domain from the Rule

No graph, no table - reason from the algebra. Ask what the rule forbids, then exclude it.

Function	What is forbidden?	Inequality	Interval notation
$f(x) = \sqrt{x - 3}$	_____	_____	_____

$g(x) = 1/(x - 5)$	_____	_____	_____
$h(x) = \sqrt{2x + 8}$	_____	_____	_____
$k(x) = 1/\sqrt{x - 1}$	_____	_____	_____

Look at $k(x)$. Why is the answer $x > 1$ and not $x \geq 1$?

At $x = 1$ the radicand is 0, and that would make the denominator _____ — so 1 must be excluded too.

Part 4 — A Domain in Three Pieces

Find the domain of $m(x) = (x + 2)/(x^2 - 9)$. Show the factoring.

Denominator factors as _____, so x cannot be _____

Domain, interval notation: _____

Domain, set notation: _____

Which notation communicates this domain more efficiently, and why?

Part 5 — Going Further

1. Explain why ∞ can never take a bracket, no matter the function.

2. Write a function whose domain is exactly $[4, \infty)$. _____

3. Write a function whose domain is exactly $(-\infty, 0) \cup (0, \infty)$. _____

4. Can a domain be empty? Give a rule with no valid inputs, or argue that none exists.

Sample: _____

Exit Ticket

Give the domain of $f(x) = \sqrt{x + 5}/(x - 2)$ in interval notation, and justify each restriction.

Domain _____

Two reasons: _____

Domain and Range from Problem Situations

Algebra II Honors | Unit 1 | Day 2 | TEKS A2.7.I, A2.2.A

Objective: I can build a function from a described situation, determine the domain the situation allows versus the domain the equation allows, and interpret both in context.

Vocabulary

Function domain — every input the _____ accepts. **Contextual domain** — every input the _____ allows.

Constraint — a condition the situation imposes that the algebra does not, such as "a length must be _____."

Average rate of change from a to b = _____ — the slope between two points.

Part 1 — Build the Function First

You are given the situation, not the equation. Write the rule, then the domain.

- a. A rectangle has perimeter 40 ft. One side is x . Write its area as a function of x .

Other side = _____ $A(x)$ = _____ Contextual domain = _____

- b. Why is the domain open at both ends rather than closed? _____

- c. What domain does the equation $x(20 - x)$ accept on its own? _____

Part 2 — Three Situations

Write the rule, the contextual domain, and the range. Use interval or set notation as fits.

1. A taxi charges \$4 plus \$2.50 per mile, for trips up to 12 miles.

$C(m)$ = _____ Domain _____ Range _____

2. A club sells shirts for \$15. Each shirt costs them \$6, and the screen fee is \$90 total.

Profit $P(n)$ = _____ Break-even at n = _____

Domain _____ Discrete or continuous? _____

3. A ball is thrown upward: $h(t) = -16t^2 + 48t + 4$ feet, until it lands.

Domain: from $t = 0$ until $h = 0$, about _____ Why can t not be negative?

The range starts at $h =$ _____ and the maximum is at $t = 1.5$, giving $h =$ _____, so range

Part 3 — Average Rate of Change

Use the taxi function $C(m) = 4 + 2.5m$ and the ball function $h(t)$ from above.

- a. Average rate of change of C from $m = 2$ to $m = 8$:

= _____ Why is this the same for every interval? _____

- b. Average rate of change of h from $t = 0$ to $t = 1$, then from $t = 2$ to $t = 3$:

First: _____ Second: _____ What does the sign change mean?

Part 4 — Going Further

1. Describe a real situation whose contextual domain is exactly the whole numbers 0 to 30.

2. Give a situation where the range is bounded but the domain is not.

Sample: _____

3. Can a contextual domain ever be LARGER than the function domain? Explain.

No — _____

Exit Ticket

A square sheet of metal 24 cm on a side has a square of side x cut from each corner. Write the volume of the resulting open box and give the contextual domain, with justification.

$V(x) =$ _____ Domain _____

Justify the upper bound: _____

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Continuous vs. Discrete

Algebra II Honors | Unit 1 | Day 3 | TEKS A2.2.A, A2.7.I

Objective: I can classify a domain as continuous or discrete, write each with the notation that fits, recognize step behavior, and judge when discreteness is a property of the quantity versus a modeling choice.

Vocabulary

Continuous — every value between any two points belongs to the domain. **Discrete** — the domain is _____.

Step function — a _____ domain paired with a _____ range. The output jumps.

Sequence — a function whose domain is the _____. Always discrete.

Part 1 — Classify and Justify

Write C or D, then give the domain in correct notation and one sentence of justification.

Situation	C/D	Domain	Why
Distance a car travels in 3 hours	_____	_____	_____
Number of students on 6 buses	_____	_____	_____
Mass of water in a beaker	_____	_____	_____
Term number in a sequence	_____	_____	_____

Part 2 — One Equation, Two Domains

A cube-shaped box of side x has volume $V(x) = x^3$.

x	1	2	3	4
V(x)	1	8	27	64

Any size box: domain _____ range _____

Built from 1-inch blocks: domain _____ range _____

The algebra did not change. What did? _____

Part 3 — Step Behavior

A parking garage charges \$3 for each hour or any part of an hour, up to 4 hours. Park for 1.2 hours and you pay for 2.

t (hours)	0.5	1.0	1.2	2.0	2.7	3.5
Cost						

Domain (time parked): _____ Range (cost): _____

This is the case the on-level notes do not cover: a domain and range that disagree. Explain how that happens.

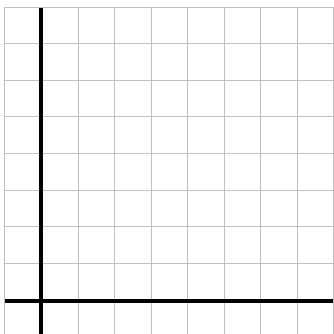
Time varies smoothly, but the _____ at each whole hour instead of rising gradually.

Why must $t = 1.0$ cost \$3 and not \$6? _____

Part 4 — Graph Both Kinds

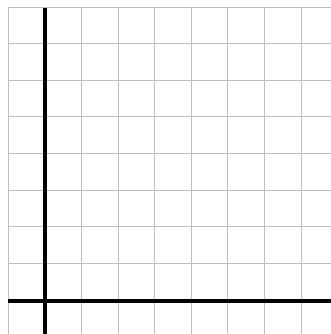
Each square is 1 unit. Use open and closed dots carefully.

- A. Discrete.** A tutor charges \$2 per session, up to 4 sessions. **B. Continuous.** A candle burns 2 cm per hour from 8 cm.



Domain _____

Range _____



Rule $h =$ _____

Domain _____ Range _____

Part 5 — Going Further

1. Is money continuous or discrete? Argue both sides, then commit to one.

Discrete in practice _____, often modeled as continuous because the steps are tiny.

2. Give a situation with a discrete domain and a continuous range, or argue it is impossible.

Impossible — _____

Exit Ticket

Postage costs \$0.60 for the first ounce and \$0.24 for each additional ounce or part of an ounce, up to 5 ounces.

Domain (weight) _____ Range (cost) _____

Classify the domain and the range, and explain why they differ: _____

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Parent Functions I: $\sqrt{x}$ and $|x|$

Algebra II Honors | Unit 1 | Day 4 | TEK A2.2.A

Objective: I can graph and analyze the square root and absolute value parent functions, write absolute value as a piecewise rule, and explain why the two behave differently for negative inputs.

Vocabulary

Reflectional symmetry — the graph can be _____ across a line and land exactly on itself.

x-intercept(s) — where the graph crosses the x-axis, where $y =$ _____. **y-intercept(s)** — where $x =$ _____.

Zeros — the x-value(s) that make a function equal _____ — the same locations as the x-intercepts.

Parent function — the simplest member of a family, with nothing added or multiplied.

Piecewise rule — one function defined by _____ on different parts of the domain.

Principal square root — $\sqrt{a}$ means the _____ root only, which is why $\sqrt{x}$ is a function.

Part 1 — Build Both Tables

Fill both. Note where outputs are undefined and where they repeat.

x	-4	-1	0	1	4	9
$f(x) = \sqrt{x}$						

x	-4	-1	0	1	4	9
$g(x) = x $						

One table has undefined entries, the other has repeats. Explain what each tells you:

$\sqrt{x}$ has a restricted _____; $|x|$ is not _____, which will matter on Day 12.

Part 2 — Absolute Value as a Piecewise Rule

Absolute value is distance from zero, so it never returns a negative. Written as two cases:

$|x| =$ _____ when $x \geq 0$, and $|x| =$ _____ when $x < 0$.

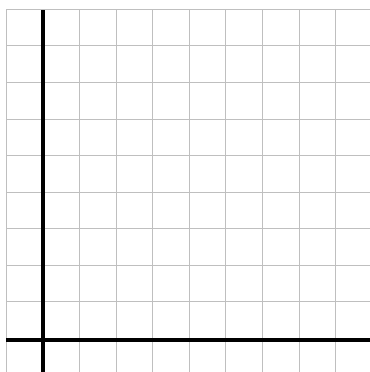
The second case looks like it produces a negative. Explain why it does not:

If x is already negative, _____. For example $-(-5) = 5$.

Part 3 — Graph Them Both

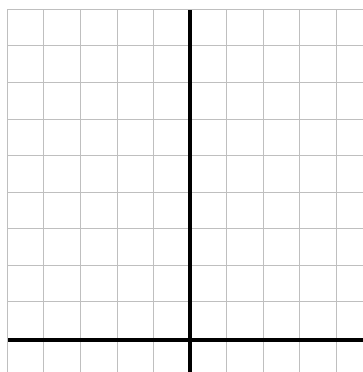
Each square is 1 unit. Plot from your tables, then connect smoothly.

A. $f(x) = \sqrt{x}$ x from 0 to 9



Domain _____ Range _____

B. $g(x) = |x|$ x from -5 to 5



Domain _____ Range _____

Part 4 — Solution Counts

Solve each. The number of answers is the point, not just the answers.

1. $|x| = 6 \rightarrow x =$ _____ how many? _____ 2. $\sqrt{x} = 6 \rightarrow x =$ _____ how many? _____

3. $|x| = 0 \rightarrow x =$ _____ how many? _____ 4. $|x| = -3 \rightarrow$ _____

Item 4 has no solution. What does that say about the range of $|x|$? _____

Connect item 1 to the graph: why do two x -values give the same output?

The V is symmetric about the _____, so a horizontal line meets it twice.

Part 5 — Going Further

1. Give the domain of each, in interval notation.

$\sqrt{x-7}$ _____ $\sqrt{3x+6}$ _____ $|x-7|$ _____

2. Why does the absolute value one have no restriction while the radicals do?

3. On the interval $[1, 9]$, give the minimum and maximum of $\sqrt{x}$. min _____ max _____

4. Is $\sqrt{x^2}$ the same function as $|x|$? Test $x = -5$ and $x = 5$, then decide.

Yes — _____

Exit Ticket

Both parent functions have minimum 0 and pass through the origin.

Name one attribute that separates them: _____ and one they share besides those two: _____

Write $|x|$ as a piecewise rule from memory: _____

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Parent Functions II: x^3 and $\sqrt[3]{x}$

Algebra II Honors | Unit 1 | Day 5 | TEK A2.2.A

Objective: I can graph and analyze the cubic and cube root parent functions, prove algebraically that each is an odd function, and describe how their rates of change differ.

Vocabulary

Odd function — satisfies _____ for every x . Its graph has _____ symmetry about the origin.

Even function — satisfies _____. Its graph is symmetric across the _____.

Average rate of change on $[a, b]$ = _____

Part 1 — Two Tables, One Relationship

x	-8	-2	-1	0	1	2	8
$f(x) = x^3$							

x	-8	-2	-1	0	1	2	8
$g(x) = \sqrt[3]{x}$							

Find a pair of entries showing the two functions undo each other: _____

Part 2 — Prove Both Are Odd

Substitute $-x$ into each rule and simplify. Do not just check one number.

a. $f(x) = x^3$. Find $f(-x)$:

$f(-x) =$ _____ so f is _____

b. $g(x) = \sqrt[3]{x}$. Find $g(-x)$:

$g(-x) =$ _____ so g is _____

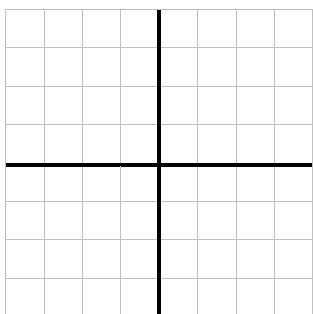
c. Test $h(x) = x^2$ the same way. Is it odd, even, or neither? _____

d. What does "odd" tell you about the graph without graphing it?

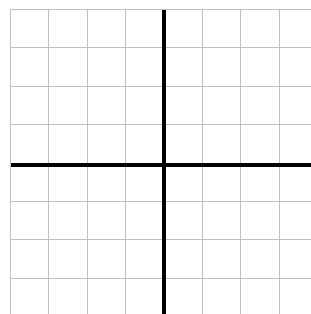
Part 3 — Graph Them Both

Both grids run -8 to 8 on each axis, so 1 square = 2 units.

A. $f(x) = x^3$



B. $g(x) = \sqrt[3]{x}$



Domain _____ Range _____

Mirror line between them: _____

Part 4 — Rates of Change

Compute the average rate of change of $f(x) = x^3$ on each interval.

On $[0, 1]$: _____ On $[1, 2]$: _____ On $[2, 3]$: _____

Now $g(x) = \sqrt[3]{x}$ on $[0, 1]$: _____ On $[1, 8]$: _____

Describe the contrast in one sentence:

The cubic _____ as x grows while the cube root _____ — which is what makes them mirror images.

Part 5 — Going Further

1. Solve $x^3 = 125$ and $x^2 = 25$. Count the real solutions for each.

$x^3 = 125 \rightarrow$ _____ $x^2 = 25 \rightarrow$ _____

2. Explain the difference using odd and even.

x^3 is odd and never repeats an output; x^2 is even, so _____

3. Which of the four parent functions so far accept negative inputs? _____

4. Can a function be both even and odd? Find one or prove none exists.

Yes — _____

Exit Ticket

A function has domain and range all reals, passes through the origin, and satisfies $f(-x) = -f(x)$.

Name two parent functions this could be: _____

Prove your choice is odd, algebraically: _____

Name _____ Date _____ Period _____

Parent Functions III: $1/x$ and Asymptotes

Algebra II Honors | Unit 1 | Day 6 | TEK A2.2.A

Objective: I can graph the reciprocal parent function, describe its end behavior precisely, prove it is odd, and locate vertical asymptotes of related rules from the denominator.

Vocabulary

Asymptotic behavior — how a graph behaves near a value it gets closer and closer to but never _____.

Vertical asymptote — a vertical line the graph _____, located where the function is _____.

Horizontal asymptote — describes _____. The graph gets closer and closer to it, without _____.

End behavior notation — "as $x \rightarrow \infty$, $f(x) \rightarrow 0$ " is read: as x grows _____, the output approaches 0.

Part 1 — Build the Table

x	-4	-2	-1	-1/2	0	1/2	1	2	4
$f(x) = 1/x$									

Why is $x = 0$ excluded, and what is the domain in set notation?

Division by zero is undefined, so _____

Part 2 — End Behavior, Precisely

Complete both patterns, then write the behavior in words.

x	1	0.1	0.01	0.001
$1/x$				

x	1	10	100	1000
$1/x$				

As $x \rightarrow 0$ from the right, $f(x) \rightarrow$ _____ As $x \rightarrow 0$ from the left, $f(x) \rightarrow$ _____

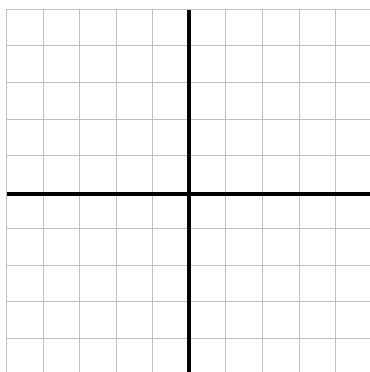
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____ As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

So the asymptotes are $x =$ _____ and $y =$ _____. Does the graph ever reach $y = 0$? _____

Part 3 — Graph and Prove

Each square is 1 unit. Two branches - do not connect them across the y -axis.

$f(x) = 1/x$, with both asymptotes dashed.



Prove f is odd.

$f(-x) =$ _____

so $f(-x) =$ _____

Quadrants used: _____

Intercepts: _____

Max or min: _____

Why does having no intercepts follow from the asymptotes?

The axes _____ the asymptotes, and a graph never reaches its asymptotes.

Part 4 — Asymptotes of Related Rules

Reason from the denominator. Where it is zero, there is a vertical asymptote.

Function	Denominator = 0 when	Vertical asymptote(s)	Domain
$1/(x - 3)$	_____	_____	_____
$1/(x^2 - 4)$	_____	_____	_____
$1/(x^2 + 1)$	_____	_____	_____

The last row has no vertical asymptote at all. Explain why:

$x^2 + 1$ is _____, so it is never zero for a real x .

Part 5 — Going Further

1. True or false, with a reason: a graph may cross a horizontal asymptote. _____

2. Does $1/x$ itself ever cross $y = 0$? _____ Is that a contradiction with item 1?

3. Solve $1/x = x$. Then say what the solutions mean on the graph.

$x =$ _____ — _____

Exit Ticket

Give the domain and both asymptotes of $f(x) = 1/(x + 5)$, and state its end behavior as $x \rightarrow \infty$.

Domain _____ VA _____ HA _____

End behavior: as $x \rightarrow \infty$, $f(x) \rightarrow$ _____

Name _____ Date _____ Period _____

Parent Functions IV: b^x and log base b

Algebra II Honors | Unit 1 | Day 7 | TEK A2.2.A

Objective: I can graph exponential and logarithmic parent functions, analyze their attributes, convert between exponential and logarithmic form, and explain why their domains and ranges are exchanged.

Vocabulary

Exponential function — the variable sits in the _____. **Base** — $b = 2, 10$, or _____.

Logarithm — log base b of a answers: what _____ on b gives a? So a logarithm IS an exponent.

Two forms — $b^x = a$ and log base b of $a = x$ say the _____ two ways.

Part 1 — Two Tables That Trade Places

x	-2	-1	0	1	2	3
$f(x) = 2^x$						

x	1/4	1/2	1	2	4	8
$g(x) = \log_2(x)$						

State precisely what happened between the two tables: _____

Can 2^x ever be 0 or negative? _____ So what is the domain of $\log_2$? _____

Part 2 — Converting Between Forms

Rewrite each in the other form. A logarithm is just an exponent asked for backwards.

Exponential form	Logarithmic form	Exponential form	Logarithmic form
$2^x = 8$	_____	$10^x = 1000$	_____
_____	log base 5 of 25 = 2	_____	$\log_2(1/8) = -3$
$3^x = 81$	_____	$e^x = 1$	_____

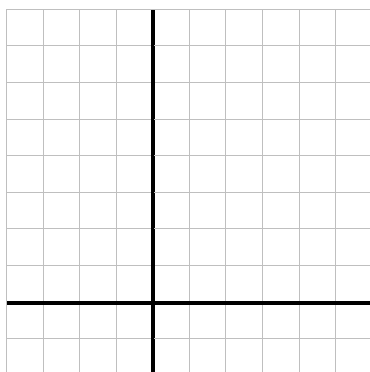
Solve by inspection: $2^x = 32 \rightarrow x =$ _____ $\log_2(64) =$ _____ $\log(100) =$ _____

Why is log base b of 1 always 0, for any base? _____

Part 3 — Graph Them Both

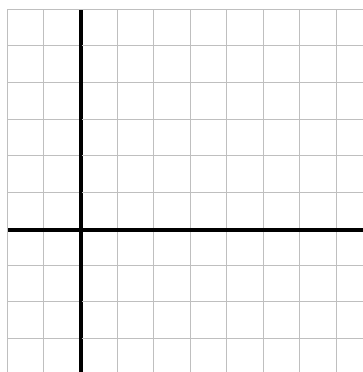
Each square is 1 unit. Draw the asymptote dashed first.

A. $f(x) = 2^x$



HA _____ intercept _____

B. $f(x) = \log_2(x)$



VA _____ intercept _____

Part 4 — Attributes Side by Side

Attribute	b^x	log base b
Domain	_____	_____
Range	_____	_____
Intercept	_____	_____
Asymptote	_____	_____

Every row is swapped. Explain what relationship between the two functions causes that:

They are _____ — formalized on Day 10.

Part 5 — Going Further

1. Which grows faster past $x = 1$, 2^x or 10^x ? Justify with two values.

2. Every exponential graph b^x passes through one shared point. Name it and explain.

3. Why is log base b of a negative number undefined?

No exponent on a positive base _____

Exit Ticket

A function has domain $(0, \infty)$, range all reals, x-intercept $(1, 0)$, and a vertical asymptote at $x = 0$.

Family: _____ Inverse family: _____ Rewrite $\log_2(16) = 4$ in exponential form: _____

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Comparing All Seven Parent Functions

Algebra II Honors | Unit 1 | Day 8 | TEK A2.2.A

Objective: I can compare all seven parent functions across their key attributes, classify each as even, odd, or neither using an algebraic test, and state end behavior in formal notation.

Vocabulary

Even: _____ **Odd:** _____ Most functions are _____.

Relative max / min — the largest or smallest output _____, which may exist even when no overall one does.

Part 1 — The Master Table

Complete every cell from your Day 4-7 work. This is your reference for the year.

Function	Domain	Range	Intercept	Asymptotes
$\sqrt{x}$	_____	_____	_____	_____
$ x $	_____	_____	_____	_____
x^3	_____	_____	_____	_____
$\sqrt[3]{x}$	_____	_____	_____	_____
$1/x$	_____	_____	_____	_____
b^x	_____	_____	_____	_____
log base b	_____	_____	_____	_____

Part 2 — Even, Odd, or Neither

Substitute $-x$ into each rule. Do not decide from the picture alone.

Function	$f(-x)$ simplifies to	Even, odd, or neither
$ x $	_____	_____
x^3	_____	_____
$\sqrt[3]{x}$	_____	_____
$1/x$	_____	_____
$\sqrt{x}$	_____	_____
2^x	_____	_____

$\sqrt{x}$ is "neither" for a different reason than 2^x . Explain the difference:

$\sqrt[3]{-x}$ is _____ for positive x , while 2^x is defined but _____.

Part 3 — Name That Function

- Odd, no intercepts, two asymptotes. _____
- Even, minimum 0, domain all reals. _____
- Domain $(0, \infty)$, range all reals, through $(1, 0)$. _____

4. Odd, domain and range all reals, and steepens as x grows. _____

5. Items 4 and its partner $\sqrt[3]{x}$ are both odd with the same domain and range. What single piece of information separates them?

One point, such as _____

Part 4 — Extremes on an Interval

a. $f(x) = x^3$ on $[-2, 2]$: min _____ max _____ b. $f(x) = |x|$ on $[-3, 1]$: min _____
max _____

c. $f(x) = 1/x$ on $[1, 4]$: min _____ max _____ d. $f(x) = 1/x$ on $[-1, 1]$:

Item (d) is the interesting one. Why does a closed interval fail to guarantee a max and min here?

The function is _____ — there is a break at $x = 0$.

Part 5 — Going Further

1. Sort all seven by whether the domain is restricted. restricted: _____

2. Only two of the seven have a minimum. Name them and explain what they share.

$\sqrt{x}$ and $|x|$ — _____

3. Must an even function pass through the origin? Give an example or counterexample.

No — _____

Exit Ticket

Classify $f(x) = 1/x$ as even, odd, or neither, showing the algebra, and name one other parent function in the same class.

$f(-x) =$ _____ Another odd one: _____

Name _____ Date _____ Period _____

Modeling with a Parent Function: the Open Box

Algebra II Honors | Unit 1 | Day 9 | TEKS A2.2.A, A2.7.I

Objective: I can model a situation with a cubic function, justify its contextual domain, locate a maximum numerically, and generalize the model to any rectangular sheet.

The Problem

A flat sheet of cardboard is **20 in. by 10 in.** A square of side x is cut from each corner and the flaps are folded up to form an open-top box.

Part 1 — Build and Expand

Each cut removes x from BOTH ends of a side. Write each dimension, then the volume.

height _____ length _____ width _____

$V(x) =$ _____

Now expand to standard form. Show the multiplication.

$V(x) =$ _____ Degree _____ Family _____

What are the three zeros of the expanded form, and which are physically meaningful?

Zeros at $x =$ _____ meaningful: _____

Part 2 — Justify the Domain

Why $x > 0$? _____

Why $x < 5$ and not $x < 10$? _____

Contextual domain _____ Open or closed, and why? _____

Part 3 — Locate the Maximum

Compute $V(x)$. Round to the nearest hundredth where needed.

x	1	2	2.1	2.11	2.2	3	4
$V(x)$							

Average rate of change on $[1, 2]$: _____ on $[2, 3]$: _____

The sign flips between those intervals. What does that tell you, and why is it stronger evidence than just reading the largest table value?

The volume stops rising and starts falling, so a maximum lies _____.

Maximum $\approx$ _____ at $x \approx$ _____ Box $\approx$ _____

Range on this domain: _____ Why is 0 excluded but 192.45 included?

Part 4 — Generalize

Now let the sheet be **L by W** with W the shorter side, and squares of side x cut from the corners.

$V(x) =$ _____ Contextual domain _____

Explain why the upper bound depends on W and never on L:

The shorter side reaches zero first, so it _____.

Apply it: a 16 by 8 sheet gives domain _____ a 30 by 30 sheet gives domain _____

Part 5 — Going Further

1. For the 16 by 8 sheet, compute $V(1)$ and $V(2)$. Show the arithmetic.

$V(1) =$ _____ $V(2) =$ _____

2. For a square sheet s by s , write $V(x)$ and its domain. $V(x) =$ _____ domain _____

3. Does doubling both sheet dimensions double the maximum volume? Predict, then justify.

No — _____

4. A sheet is 18 by 6. Find the domain, then compute V at the midpoint of that domain.

Domain _____ midpoint $x =$ _____ $V(1.5) =$ _____

5. Is the maximum always at the midpoint of the domain? Test against the 20 by 10 sheet.

No — _____

Exit Ticket

A 12 in. by 12 in. sheet has squares of side x cut from each corner.

$V(x) =$ _____ Domain _____ $V(2) =$ _____

Explain the upper bound in one sentence: _____

Inverses from Graphs and Tables

Algebra II Honors | Unit 1 | Day 10 | TEK A2.2.B

Objective: I can find the inverse of a relation from a table or graph, use $f^{-1}(x)$ notation correctly, justify why domain and range exchange, and predict whether an inverse will be a function.

Vocabulary

- Inverse** — the relation that _____ the original. Written _____, which does _____ mean $1/f(x)$.
- Building it** — in a table or a set of pairs, _____. On a graph, reflect across _____.
- One-to-one** — no output is repeated. Test with a _____ line: if it meets the graph twice, the function is not one-to-one.

Part 1 — Inverses from Pairs and Tables

- a. $f = \{ (-4, -6), (0, -1), (2, 4), (3, -7), (7, 7) \}$
 $f^{-1} = \{ \text{_____} \}$
- b. Complete the inverse table.

x	-4	0	2	3	7
f(x)	-6	-1	4	-7	7

x	-6	-1	4	-7	7
f ⁻¹ (x)					

- c. Which point in f is unchanged by the swap, and what is special about it? _____

Part 2 — Why Domain and Range Exchange

- Argue this in general, not from one example.
- If (a, b) belongs to f , then _____ belongs to f^{-1} by definition of the swap.
- So every input of f^{-1} was an _____ of f . Therefore:
- domain of $f^{-1} =$ _____ range of $f^{-1} =$ _____

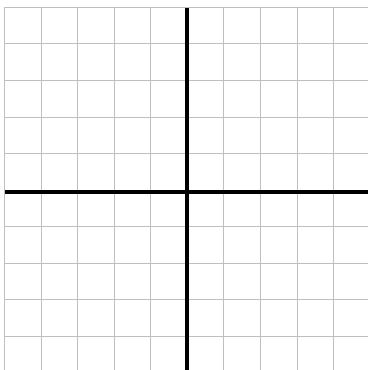
f	Domain of f	Range of f	Domain of f ⁻¹	Range of f ⁻¹
x^3	all reals	all reals	_____	_____
2^x	all reals	$(0, \infty)$	_____	_____
$\sqrt{x}$	$[0, \infty)$	$[0, \infty)$	_____	_____

Row 2 explains something from Day 7. What? _____

Part 3 — Graph a Pair

Each square is 1 unit. Plot f , then f^{-1} , then draw $y = x$.

$$f = \{ (-4, -2), (0, 0), (1, 1), (2, 4) \}$$



Record it.

Reflection line: _____

Fixed points: _____

If (a, b) is on f , then on f^{-1} :

Is f one-to-one here? _____

$$f^{-1} = \{ \text{_____} \}$$

Part 4 — Will the Inverse Be a Function?

Apply the horizontal line test to the original, before building the inverse.

a. $g = \{ (2, 6), (4, 6), (5, 1) \}$. Write g^{-1} : _____

Is g^{-1} a function? _____ Why? _____

What was true of g that caused it? _____

b. Which of these parent functions have inverses that are functions? Circle:

x^3 x^2 2^x $|x|$ $\sqrt[3]{x}$ Answer: _____

What do the ones that fail have in common? _____

Part 5 — Going Further

1. Does $f^{-1}(x)$ mean $1/f(x)$? Give a specific counterexample with numbers.

For $f(x) = x + 3$: $f^{-1}(x) = \text{_____}$ but $1/f(x) = \text{_____}$ — different functions.

2. A function is its own inverse. What must be true of its graph? _____

3. Can a function have two different inverses? Argue yes or no.

No — _____

Exit Ticket

For $h = \{ (1, 4), (2, 9), (5, 0) \}$, write h^{-1} , give its domain and range, and say whether it is a function with a reason.

$h^{-1} = \{ \text{_____} \}$ Domain _____ Range _____

A function? _____

Name _____ Date _____ Period _____

Writing the Inverse of a Linear Function

Algebra II Honors | Unit 1 | Day 11 | TEK A2.2.B

Objective: I can write the inverse of a linear function algebraically, derive the inverse of the general form $f(x) = mx + b$, and prove that a line and its inverse have reciprocal slopes.

Warm-Up — This Is a Skill You Have

Finding an inverse is solving a literal equation for the other variable. Nothing new.

Solve $y = 3x + 6$ for x : $x =$ _____ Solve $y = mx + b$ for x : $x =$ _____

The Procedure

1. Replace $f(x)$ with _____.
2. _____.
3. Solve for y and rename it _____.

Part 1 — Find Each Inverse

Show the swap-and-solve steps, not just the answer.

a. $f(x) = 2x + 3$

$f^{-1}(x) =$ _____

b. $f(x) = -2x + 8$

$f^{-1}(x) =$ _____

c. $f(x) = (1/2)x - 6$

$f^{-1}(x) =$ _____

d. $f(x) = (3x - 1)/4$

$f^{-1}(x) =$ _____

Part 2 — Derive the General Case

Now do it once for every linear function at the same time. Let $f(x) = mx + b$, with $m \neq 0$.

$f^{-1}(x) =$ _____ which in slope-intercept form is $y =$ _____

Slope of f is _____ and slope of f^{-1} is _____, so the slopes are _____. That is now proved, not just noticed.

Why did the derivation require $m \neq 0$? What kind of line is $m = 0$, and what goes wrong?

A _____ line. Every input maps to one output, so the inverse _____.

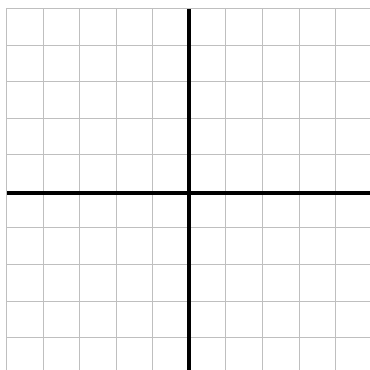
Part 3 — Check Two Ways

Use $f(x) = 2x + 3$ and $f^{-1}(x) = (x - 3)/2$.

x	-1	0	1	2
f(x)	1	3	5	7

x	1	3	5	7
f⁻¹(x)				

Graph f , f^{-1} , and $y = x$.



Record it.

slope of f _____ slope of f^{-1} _____

y-int of f _____ of f^{-1} _____

Where do f and f^{-1} intersect?

Part 4 — Self-Inverse Lines

A function is self-inverse when $f^{-1}(x) = f(x)$. Using $f(x) = mx + b$, find every linear function with this property.

Need $(x - b)/m = mx + b$, which forces $m =$ _____

If $m = 1$, b must be _____, giving _____. If $m = -1$, b may be _____, giving $f(x) = -x + b$.

Check one: is $f(x) = -x + 6$ its own inverse? Test with $x = 2$.

$f(2) =$ _____ $f(4) =$ _____ so yes, it returns the start. _____

Part 5 — In Context

Celsius to Fahrenheit: $F(c) = (9/5)c + 32$.

$F^{-1}(x) =$ _____ In words, the inverse _____

Is there a temperature where the two scales agree? Set $(9/5)c + 32 = c$ and solve.

$c =$ _____ — _____

Exit Ticket

Write the inverse of $f(x) = 6x - 12$, verify with one input, and state the relationship between the two slopes.

$f^{-1}(x) =$ _____ $f(3) =$ _____ and $f^{-1}(6) =$ _____

Slopes: _____

Name _____ Date _____ Period _____

When Is an Inverse a Function?

Algebra II Honors | Unit 1 | Day 12 | TEK A2.2.C

Objective: I can determine whether the inverse of a function is a function, choose a domain restriction that makes it one, and apply this to a quadratic whose vertex is not at the origin.

Vocabulary

One-to-one — every output comes from exactly one input. Detected by the _____.

Key fact — the inverse of a function is always a _____, but a _____ only when the original is one-to-one.

Domain restriction — deliberately shrinking the domain to force one-to-one behavior. Restricting the domain of f restricts the _____ of f^{-1} .

Part 1 — Why $f(x) = x^2$ Fails

x	-3	-2	-1	0	1	2	3
$f(x) = x^2$							

Repeated outputs: _____ So f is _____

Swapping produces both $(1, 1)$ and $(1, -1)$. State precisely which test the inverse fails and why:

The _____ — the input 1 would need two outputs.

Solving $y = x^2$ for x gives $x = \pm \sqrt{y}$, and the _____ is the whole problem.

Part 2 — Restrict and Repair

Restriction on x^2	Half of the parabola kept	Resulting $f^{-1}(x)$	Range of f^{-1}
$x \geq 0$	_____	_____	_____
$x \leq 0$	_____	_____	_____

Both restrictions work. Is one more correct than the other? _____

Part 3 — A Vertex That Is Not at the Origin

Now take $g(x) = (x - 2)^2$. The same idea applies, but you must locate the turning point first.

Vertex of g : _____ So the natural restriction is _____

Now find the inverse. Swap and solve for y , keeping track of the $\pm$.

$x = (y - 2)^2 \rightarrow \pm \sqrt{y} + 2 = y \rightarrow y = \pm \sqrt{x - 4} + 2$

With the restriction $x \geq 2$ on g , the inverse is $g^{-1}(x) = \sqrt{x - 4} + 2$ Domain of g^{-1} _____

Why does the 2 appear OUTSIDE the radical in the inverse when it was inside the square in g?

Undoing happens in _____: take the root first, then add 2.

Part 4 — Which Need No Restriction?

Function	One-to-one?	Restriction needed?	Inverse
x^2	_____	_____	_____
x^3	_____	_____	_____
2^x	_____	_____	_____
$ x $	_____	_____	_____
$1/x$	_____	_____	_____

State the pattern as a rule about increasing and decreasing:

A function that only increases or only decreases is _____; one that turns around is not.

Part 5 — Going Further

1. Restrict $h(x) = (x + 3)^2$ so its inverse is a function, then write the inverse.

Restrict _____ $h^{-1}(x) =$ _____

2. Explain why $1/x$ is its own inverse, using the swap.

3. Is there a restriction of $|x|$ whose inverse is a function? Give one. _____

Exit Ticket

Consider $f(x) = (x - 5)^2$ with no restriction.

Is its inverse a function? _____ Give a restriction that fixes it: _____

With that restriction, $f^{-1}(x) =$ _____ Explain in one sentence why the restriction is needed:

Name _____ Date _____ Period _____

Composition to Verify Inverses

Algebra II Honors | Unit 1 | Day 13 | TEK A2.2.D

Objective: I can compose two functions, use $f(g(x)) = x$ and $g(f(x)) = x$ to prove they are inverses, and state the domain restriction a non-linear pair requires.

Vocabulary

Composition — one function inside another, $f(g(x))$ or $(f \circ g)(x)$. Evaluate from the _____.

The inverse test — f and g are inverses only when _____ AND _____. One direction is not enough.

Order matters — in general $f(g(x))$ _____ the same as $g(f(x))$.

Part 1 — Warm Up and Prove

Let $f(x) = 3x - 5$ and $g(x) = (x + 5)/3$.

$f(4) =$ _____ $g(7) =$ _____ $f(g(4)) =$ _____ $g(f(4)) =$ _____

Now prove it in general, both directions.

$f(g(x)) = 3 \cdot (x+5)/3 - 5 =$ _____ $g(f(x)) = ((3x-5)+5)/3 =$ _____ Inverses? _____

Show that order matters in general: for $f(x) = x + 1$ and $h(x) = 2x$, find both compositions.

$f(h(x)) =$ _____ $h(f(x)) =$ _____ Equal? _____

Part 2 — A Pair That Looks Right and Is Not

Test $f(x) = 2x + 1$ and $g(x) = 2x - 1$. Compute, do not assume.

$f(g(x)) =$ _____ Equal to x ? _____ Actual inverse of f : _____

State the lesson in your own words:

Changing a sign is not undoing a function — you must reverse _____.

Part 3 — Non-Linear Pairs and Restrictions

Test $f(x) = x^2$ and $g(x) = \sqrt{x}$. Try $x = 9$ and then $x = -9$.

$f(g(9)) =$ _____ $g(f(9)) =$ _____ $g(f(-9)) =$ _____

So $g(f(x)) = x$ holds only when _____. In general $g(f(x)) = \sqrt{x^2} =$ _____, not x .

Write the complete, correct statement of the inverse relationship for this pair:

f and g are inverses _____

Now test $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$ the same way, at $x = -8$.

$g(f(-8)) =$ _____ Restriction needed? _____ Why the difference from the squaring pair?

Part 4 — Error Analysis

A student is given $f(x) = 2(x - 3)^2 + 4$ and claims $f^{-1}(x) = \sqrt{x - 3} + 4$.

Test the claim at $x = 6$: the claimed inverse gives _____, and f of that is _____.

Undo the steps in reverse: subtract 4, divide by 2, take the root, add 3.

Correct: $f^{-1}(x) =$ _____ Restriction making it a function: _____

Where does the $\pm$ come from, in terms of Day 12? _____

Part 5 — Going Further

1. Verify $f(x) = 5x$ and $g(x) = x/5$ are inverses, both directions. _____

2. Show $f(x) = 1/x$ is its own inverse by composing it with itself.

$f(f(x)) = 1/(1/x) =$ _____ Other self-inverse functions: _____

3. Prove in general that $f(x) = mx + b$ and $f^{-1}(x) = (x - b)/m$ compose to x.

4. If $f(g(x)) = x$ but $g(f(x)) \neq x$, can they still be called inverses? Explain.

No — _____

Exit Ticket

Decide whether $f(x) = 4x - 8$ and $g(x) = (x + 8)/4$ are inverses. Show both compositions, then explain why one direction would not be sufficient proof.

$f(g(x)) =$ _____ $g(f(x)) =$ _____ Inverses? _____